Pollutants

SWAT+ Documentation

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Purpose

SWAT+ simulates the pesticides listed in `constituents.cs`. It follows each one on plant foliage, in the soil layers, in surface runoff, lateral flow, tile flow and percolation, in channel water and bed sediment, in reservoirs and in aquifers. Four files describe them:

File	Named in	Content
`pesticide.pes`	`hru_parm_db` line of `file.cio` (default `pesticide.pes`)	Properties of each pesticide.
`pest_hru.ini`	`init` line of `file.cio`	Initial pesticide in soils and on plants.
`pest_water.ini`	`init` line of `file.cio`	Initial pesticide in channels, reservoirs and aquifers.
`pest_metabolite.pes`	fixed name, not in `file.cio`	Degradation products (daughters) of each parent pesticide.

Every pesticide named in `constituents.cs` must have a row in `pesticide.pes` with the same name. The database can contain many more pesticides than the run simulates. Pesticide applications in `management.sch` and decision tables name rows of `pesticide.pes`, and the application method comes from `chem_app.ops`.

pesticide.pes

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one pesticide with 17 values and a description, in the order below. The reader does not match columns by header name, so every value must be present and in this order.

#	Column	Units	Typical range	Description
1	`name`	-	-	Name of the pesticide, up to 16 characters. Must be unique.
2	`soil_ads`	(mg/kg)/(mg/L)	1 to 10^9	Soil adsorption coefficient normalized for soil organic carbon (K_{oc}).
3	`frac_wash`	fraction	0 to 1	Fraction of the pesticide on foliage washed off by a rainfall event.
4	`hl_foliage`	days	0 to 10,000	Half-life on foliage. 0 means no decay.
5	`hl_soil`	days	0 to 100,000	Half-life in the soil. 0 means no decay.
6	`solub`	mg/L	-	Solubility in water. Upper limit of the concentration in water leaving a soil layer.
7	`aq_hlife`	days	-	Half-life in channel, reservoir and aquifer water. 0 means no decay. Earlier documentation had `aq_reac` here, a reaction rate in 1/day; the column is now a half-life in days.
8	`aq_volat`	m/day	-	Volatilization coefficient in channel and reservoir water.
9	`mol_wt`	g/mol	-	Molecular weight. Used for the water and sediment mixing velocity and for the mass of metabolites.
10	`aq_resus`	m/day	-	Resuspension velocity of pesticide sorbed to bed sediment.
11	`aq_settle`	m/day	-	Settling velocity of pesticide sorbed to suspended sediment.
12	`ben_act_dep`	m	-	Depth of the active bed sediment layer.
13	`ben_bury`	m/day	-	Burial velocity in bed sediment.
14	`ben_hlife`	days	-	Half-life in bed sediment. 0 means no decay. Earlier documentation had `ben_reac` here, a reaction rate in 1/day; the column is now a half-life in days.
15	`pl_uptake`	-	0 and up	Transpiration stream concentration factor (TSCF): ratio of the concentration in the water taken up by roots to the dissolved concentration in the soil. 0 turns plant uptake off.
16	`harvest_sink`	0 or 1	0, 1	1: pesticide taken up by the plant leaves the system. 0: it stays in the plant and returns to the soil with the residue when the plant is harvested or killed.
17	`resistance_n`	fraction	0 to 1	Soil water used for the dissolved concentration in plant uptake. 0 uses the actual soil water, 1 uses field capacity, values in between blend the two.
18	`description`	-	-	Free text, one word.

Columns 15 to 17 are new in 62.0.1. Files written for earlier revisions have 14 values (name and 13 numbers) and then a description. With such a file the first 14 values of each row are read correctly, and the read of the row stops with an error when it meets the description where `pl_uptake` is expected. No message is written and the next row is read normally. The three new columns keep their built-in values (`pl_uptake` = 0, `harvest_sink` = 1, `resistance_n` = 1), so there is no plant uptake, which is the behaviour of earlier revisions. To use plant uptake, add the three columns before the description.

Half-lives and decay

Each day the pesticide in a compartment decays by first-order kinetics. The fraction left after one day is

$$f = \exp\left(-\frac{0.693}{t_{1/2}}\right)$$

where $t_{1/2}$ is the half-life of that compartment (`hl_foliage`, `hl_soil`, `aq_hlife` or `ben_hlife`). The half-life entered is a lumped value covering volatilization, photolysis, hydrolysis, biological degradation and other reactions. In channels the decay is scaled by the travel time through the reach, up to one day.

For most pesticides the foliar half-life is much shorter than the soil half-life because of volatilization and photodecomposition. If no foliar value is available, a common estimate is 0.25 to 0.5 times the soil half-life, lower for pesticides with a vapor pressure below 10^{-5} mm Hg, and at most 30 days.

Temperature-dependent decay

Column 7 of `parameters.bsn` switches temperature-dependent decay on. The SWAT+ Editor writes this column under the header `exp_sed`. When its value is 1, each half-life is taken as the value at 25 °C and adjusted with a Q10 of 2.58:

$$t_{1/2}(T) = t_{1/2,25}\ 2.58^{(25-T)/10}$$

The temperature T is the soil layer temperature for soil, the top soil layer temperature for foliage, and the water temperature for channel water and bed sediment. For aquifers the temperature is 0 °C in 62.0.1, so aquifer half-lives are about 10.7 times the 25 °C value. Reservoirs do not use the adjustment.

Any other value in that column gives the constant decay above. Check this column when a model is moved to 62.0.1: editor files commonly have 1.0 there, which switches the adjustment on.

Sorption and transport

The soil adsorption coefficient of a layer is

$$K_d = K_{oc}\,\frac{OC}{100}$$

where OC is the organic carbon content of the layer in percent. The pesticide in a layer is split between the soil and the water flowing through it according to K_d. The concentration in water leaving a layer is limited to `solub`. The concentration in surface runoff is the concentration in percolate times `pest_perc` of `parameters.bsn`.

On a day with at least 2.54 mm of precipitation, `frac_wash` of the pesticide on each plant's foliage is washed off to the top soil layer.

Plant uptake

When `pl_uptake` is above 0, growing plants take up pesticide from each soil layer with the water they transpire:

$$U = \mathrm{TSCF}\,\frac{W_{up}}{\theta_{eff} + W_{up} + K_d\,\rho_b\,h}\,M$$

where W_{up} is the water taken up from the layer (mm), θ_{eff} is the soil water blended toward field capacity by `resistance_n`, ρ_b the bulk density, h the layer thickness (mm) and M the pesticide in the layer (kg/ha). Uptake is reported as `pl_uptake` in the pesticide output. With `harvest_sink` = 0, the pesticide kept in the plant returns to the soil when the plant is harvested or killed.

Metabolites

When a parent pesticide decays, a daughter listed in `pest_metabolite.pes` receives

$$M_{daughter} = \Delta M_{parent}\, f_{frac}\, \frac{MW_{daughter}}{MW_{parent}}$$

where f_{frac} is the daughter fraction for that compartment and MW the `mol_wt` values. Foliage uses `foliar_fr`, soil uses `soil_fr`, channel water uses `aq_fr` and channel bed sediment uses `ben_fr`. In 62.0.1 reservoirs (water and bed) and aquifers use `soil_fr`.

Calibration

These parameters can be changed during calibration through calibration.cal: `pst_koc`, `pst_washoff`, `pst_foliar_hlife`, `pst_soil_hlife`, `pst_solub`, `pst_aq_hlife`, `pst_aq_volat`, `pst_aq_settle`, `pst_aq_resus`, `pst_ben_hlife`, `pst_ben_bury` and `pst_ben_act_dep`. The daily decay factors are computed from the half-lives when `pesticide.pes` is read, so a calibration change to a half-life only has an effect when temperature-dependent decay is on.

pesticide.pes example

The first rows of `refdata/Osu_1hru/pesticide.pes` in the SWAT+ repository, as shipped:

```
pesticide.pes: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                  soil_ads     frac_wash    hl_foliage       hl_soil          solub      aq_hlife
245-tp              2600.00000       0.40000       5.00000      20.00000        2.50000       0.00700
2plus2                20.00000       0.95000      10.00000      21.00000   660000.00000       0.00700
```

The file was written for revision 60.5.4 and has no `pl_uptake`, `harvest_sink` or `resistance_n` columns. As described above, each row is read up to `ben_hlife`, the description is not stored, and the three new columns take their built-in values, so no pesticide is taken up by plants.

Silvex (`245-tp`) sorbs strongly (K_{oc} of 2600) and has a soil half-life of 20 days. Mecoprop (`2plus2`) is very soluble and hardly sorbs.

Every row has `aq_hlife` = 0.007 and `ben_hlife` = 0.05. These match the reaction coefficients in 1/day of the older database format, but 62.0.1 reads them as half-lives in days. A half-life of 0.007 days leaves a fraction $\exp(-0.693/0.007) \approx 10^{-43}$ after one day, and 0.05 days leaves about 10^{-6}. Osu_1hru has 1 in column 7 of parameters.bsn, so the half-lives are adjusted for temperature (except in reservoirs), which lengthens them at temperatures below 25 °C but not enough to change the result. Any of these pesticides that reached channel water, bed sediment, a reservoir or an aquifer would be almost entirely gone within a day. Osu_1hru has no constituents.cs, so in that model no pesticide is simulated and the file is read without effect.

pest_hru.ini

Initial pesticide in the soil and on plants of HRUs. An HRU selects a record by name in the `pest` column of soil_plant.ini.

The file is read in two passes that expect different layouts:

- The first pass counts records. It expects a title line, then for each record a header line, a name line and one line per pesticide.
- The second pass reads the values. It expects a title line and one header line, then for each record a name line and one line per pesticide.

A file with one record satisfies both. With more than one record the second pass reads the header line of record 2 as its name and the values that follow are shifted, so use one record.

Each pesticide line has three values, one line per pesticide in the order of constituents.cs:

Value	Units	Description
name	-	Pesticide name, for readability. Not checked.
soil	mg/kg	Concentration in the soil. The same concentration is given to every soil layer.
plant	kg/ha	Pesticide on the foliage. It is shared among the plants by leaf area; if there is no leaf area at the start, nothing is put on the foliage.

pest_water.ini

Initial pesticide in channels, reservoirs, wetlands and aquifers. Channels, reservoirs and wetlands select a record by name in their initial condition files (initial.cha, initial.res), and aquifers in initial.aqu.

Line 1 is a title and line 2 a header. Each record starts with a name line. The values follow as one list: a label, then the water value of each pesticide, then the bed sediment value of each pesticide, in the order of constituents.cs. Line breaks within the list are free, but the counting pass expects two lines per pesticide after the name line. For two pesticides:

```
pest_water.ini: initial pesticide in water
name
ch_init
atrazine  0.002
0.001
0.05
0.02
```

Here `atrazine` is the label, the water values are 0.002 and 0.001, and the bed sediment values 0.05 and 0.02.

Value	Units	Description
water	mg/L	Concentration in the water. Channels multiply it by the initial channel water volume. Aquifers read it as mg/kg of aquifer material over the aquifer depth with a bulk density of 2.0 t/m^3.
benthic	mg/kg	Concentration in the active bed sediment layer. Channels multiply it by the volume of the active bed layer.

Reservoirs use both values as given, without a conversion by volume.

pest_metabolite.pes

The degradation products of each parent pesticide. SWAT+ reads it if a file of this name exists in the run directory.

Line 1 is a title and line 2 a header. For each parent there is one line with the parent name and the number of daughters, followed by one line per daughter:

Value	Units	Description
`name`	-	Daughter pesticide name. Must also be listed in constituents.cs and `pesticide.pes`.
`foliar_fr`	fraction	Fraction of the parent decayed on foliage that becomes this daughter.
`soil_fr`	fraction	Fraction of the parent decayed in the soil that becomes this daughter. Also used in reservoirs and aquifers.
`aq_fr`	fraction	Fraction of the parent decayed in channel water that becomes this daughter.
`ben_fr`	fraction	Fraction of the parent decayed in channel bed sediment that becomes this daughter.

The parent name must match a row of `pesticide.pes`. If it does not, its daughter lines are not skipped and the rest of the file is read out of step.

```
pest_metabolite.pes: metabolites
parent        num_metab
aatrex        1
deethyl_atr   0.2   0.3   0.1   0.1
```

Atrazine (`aatrex`) has one daughter. 30% of the atrazine decayed in the soil, adjusted by the ratio of molecular weights, becomes `deethyl_atr`.

Related files

- constituents.cs: the list of pesticides simulated.
- soil_plant.ini: links HRUs to `pest_hru.ini` records.
- initial.cha, initial.res, initial.aqu: link channels, reservoirs and aquifers to `pest_water.ini` records.
- chem_app.ops and management.sch: pesticide applications.
- parameters.bsn: `pest_perc` and the temperature-dependent decay switch in column 7.
- Pathogens and heavy metals use the same initial condition layout.

Sources

- Readers: pest_parm_read.f90, pest_hru_aqu_read.f90, pest_cha_res_read.f90, pest_metabolite_read.f90
- Types: `type pesticide_db`, `type pesticide_cp`, `type daughter_decay_fractions` in pesticide_data_module.f90; `type cs_soil_init_concentrations` and `type cs_water_init_concentrations` in constituent_mass_module.f90
- Pesticide list crosswalk: constit_db_read.f90
- Initial conditions: pesticide_init.f90, sd_hydsed_init.f90, res_initial.f90, aqu_read_init_cs.f90
- Landscape processes: pest_decay.f90, pest_washp.f90, pest_lch.f90, pest_pl_up.f90, mgt_harvbiomass.f90, mgt_harvtuber.f90
- Temperature adjustment: pest_tempdecay_module.f90
- Channels, reservoirs and aquifers: ch_rtpest.f90, res_pest.f90, aqu_1d_control.f90
- Switch in parameters.bsn: `bsn_prm%temp_decay` in basin_module.f90
- Calibration parameters: cal_parm_select.f90

Column	Fortran variable
`name`	`pestdb%name`
`soil_ads`	`pestdb%koc`
`frac_wash`	`pestdb%washoff`
`hl_foliage`	`pestdb%foliar_hlife`
`hl_soil`	`pestdb%soil_hlife`
`aq_hlife`, `aq_volat`, `aq_resus`, `aq_settle`	`pestdb%` same names
`ben_act_dep`, `ben_bury`, `ben_hlife`	`pestdb%` same names
`pl_uptake`, `harvest_sink`, `resistance_n`	`pestdb%` same names
`description`	`pestdb%descrip`
parameters.bsn column 7 (`exp_sed`)	`bsn_prm%temp_decay`
pest_hru.ini record	`pest_soil_ini%name`, `%soil`, `%plt`
pest_water.ini record	`pest_init_name`, `pest_water_ini%water`, `%benthic`
pest_metabolite.pes daughter	`pestcp%daughter`

Purpose

SWAT+ can follow pathogens (bacteria and other microorganisms) listed in constituents.cs. Three files describe them:

File	Named in	Content
`pathogens.pth`	`hru_parm_db` line of `file.cio` (default `pathogens.pth`)	Die-off, growth and transport properties of each pathogen.
`path_hru.ini`	`init` line of `file.cio`	Initial pathogens in soils and on plants.
`path_water.ini`	`init` line of `file.cio`	Initial pathogens in channels, reservoirs and aquifers.

Every pathogen named in `constituents.cs` should have a row in `pathogens.pth` with the same name.

The pathogen code in 62.0.1 does not yet work as a complete model. Several columns are read but not used, and the landscape and channel routines have the limitations listed in Behaviour in 62.0.1. Read that section before using pathogen results.

pathogens.pth

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one pathogen with the 18 values below in this order. The reader does not match columns by header name.

#	Column	Units	Used in 62.0.1	Description
1	`name`	-	yes	Pathogen name, up to 16 characters.
2	`do_soln`	1/day	yes	Die-off factor in soil solution at 20 °C.
3	`gr_soln`	1/day	yes	Growth factor in soil solution at 20 °C.
4	`do_sorb`	1/day	no	Die-off factor for pathogens sorbed to soil particles at 20 °C.
5	`gr_sorb`	1/day	no	Growth factor for pathogens sorbed to soil particles at 20 °C.
6	`kd`	-	yes	Partition coefficient between solution and sorbed phases.
7	`t_adj`	-	yes	Temperature adjustment factor for die-off and growth.
8	`washoff`	fraction	yes	Fraction of the pathogens on foliage washed off by a rainfall event.
9	`do_plnt`	1/day	yes	Die-off factor on foliage at 20 °C.
10	`gr_plnt`	1/day	yes	Growth factor on foliage at 20 °C.
11	`fr_manure`	fraction	no	Fraction of manure containing active colony forming units.
12	`perco`	-	yes	Percolation coefficient: ratio of the concentration in percolate to the concentration in the surface layer solution.
13	`det_thrshd`	cfu/m²	no	Detection threshold below which soil pathogens would be set to zero.
14	`do_stream`	1/day	yes	Die-off factor in streams at 20 °C.
15	`gr_stream`	1/day	no	Growth factor in streams at 20 °C.
16	`do_res`	1/day	no	Die-off factor in reservoirs at 20 °C.
17	`gr_res`	1/day	no	Growth factor in reservoirs at 20 °C.
18	`conc_min`	cfu/m²	yes	Minimum pathogen level on foliage and in the soil surface layer.

Die-off and growth

On foliage and in the top soil layer the pathogen count N changes each day by the net die-off rate, adjusted for the mean air temperature T:

$$N_d = N_{d-1}\,\exp\!\left(-(k_{die} - k_{grow})\,\theta^{T-20}\right) - c_{min}$$

where k_{die} and k_{grow} are `do_plnt` and `gr_plnt` on foliage, or `do_soln` and `gr_soln` in the soil, θ is `t_adj` and c_{min} is `conc_min`. The result is not allowed to fall below `conc_min`.

On a day with at least 2.54 mm of precipitation, `washoff` of the pathogens on each plant's foliage moves to the top soil layer.

path_hru.ini

Initial pathogens in the soil and on plants of HRUs. An HRU selects a record by name in the `path` column of `soil_plant.ini`.

Line 1 is a title. Each record has a header line, a name line, and then the values as one list: a label, the soil value of each pathogen, then the plant value of each pathogen, in the order of `constituents.cs`. Line breaks within the list are free, but the list must take one line per pathogen so that the records are counted correctly. For two pathogens:

```
path_hru.ini: initial pathogens
name
hru_init
ecoli  1000.  500.
50.  10.
```

Here `ecoli` is the label, the soil values are 1000 and 500 and the plant values 50 and 10.

Value	Units	Description
soil	cfu/m²	Pathogens in the top soil layer.
plant	cfu/m²	Pathogens on the foliage.

path_water.ini

Initial pathogens in channels, reservoirs and aquifers. Reservoirs and channels select a record by name in their initial condition files (`initial.res`, `initial.cha`), and aquifers in `initial.aqu`.

Line 1 is a title and line 2 a header. The record has a name line and then the values as one list: a label, the water value of each pathogen, then the bed sediment value of each pathogen. The list must take two lines per pathogen. Use one record (see below).

Value	Units	Description
water	-	Pathogens in the water. The value is used as given, without conversion, so give it in the units of the pathogen loads in the model.
benthic	-	Pathogens in the bed sediment, used as given.

Behaviour in 62.0.1

- Columns `do_sorb`, `gr_sorb`, `fr_manure`, `det_thrshd`, `gr_stream`, `do_res` and `gr_res` are not used.
- The landscape routines (die-off, wash-off, runoff and percolation) do not look up each pathogen's row by name. They use the row of `pathogens.pth` whose position equals the position of the HRU's record in `path_hru.ini`, and use that one row for every pathogen.
- Each day the runoff step removes all pathogens left in the top soil layer after percolation and reports them as lost in surface runoff, whether or not there is runoff that day.
- Initial soil pathogens from `path_hru.ini` are set to zero again for any soil with more than one layer. Initial plant pathogens are written to the balance output but not put on the foliage.

- `path_water.ini` is read with a header line before each record in the reading pass and without one in the counting pass, so only a file with one record is read correctly.
- Channels take their initial pathogens from the `path_water.ini` record in the position of the channel's pesticide initial record, not its pathogen record. Aquifers take the name match from `path_water.ini` but the values from the `path_hru.ini` soil column of the record in the same position. Reservoirs use their `path_water.ini` record.
- The channel die-off step does not identify the channel being routed, so `do_stream` does not act on the routed water.

Related files

- `constituents.cs`: the list of pathogens simulated.
- `soil_plant.ini`: links HRUs to `path_hru.ini` records.
- `initial.cha`, `initial.res`, `initial.aqu`: link channels, reservoirs and aquifers to `path_water.ini` records.
- Pesticides and heavy metals use a similar initial condition layout.

Sources

- Readers: `path_parm_read.f90`, `path_hru_aqu_read.f90`, `path_cha_res_read.f90`
- Types: `type pathogen_db` in `pathogen_data_module.f90`; `type cs_soil_init_concentrations` and `type cs_water_init_concentrations` in `constituent_mass_module.f90`
- Pathogen list crosswalk: `constit_db_read.f90`; `soil_plant.ini` link: `hru_read.f90`
- Initial conditions: `pathogen_init.f90`, `sd_hydsed_init.f90`, `res_initial.f90`, `aqu_read_init_cs.f90`
- Landscape processes: `path_ls_swrouting.f90`, `path_ls_runoff.f90`, `path_ls_process.f90`, called from `hru_control.f90`
- Channel: `ch_rtpath.f90`; temperature function: `theta.f90`

Column	Fortran variable
`name`	`path_db%pathnm`
other pathogens.pth columns	`path_db%` same names
path_hru.ini record	`path_soil_ini%name`, `%soil`, `%plt`
path_water.ini record	`path_init_name`, `path_water_ini%water`, `%benthic`

Purpose

Heavy metals listed in `constituents.cs` can be routed through the network as loads. In 62.0.1 they enter the model only through export coefficients (`exco_hmet.exc`) and delivery ratios (`dr_hmet.del`) and are routed with the water. There is no metal property database, and soils, plants, channels and reservoirs have no metal processes.

`file.cio` has slots for four metal files:

File	Named in	Status in 62.0.1
`metals.mtl`	`hru_parm_db` line of `file.cio`	No reader. The file is not opened.
`hmet_hru.ini`	`init` line of `file.cio`	Read, but the values are not used.
`hmet_water.ini`	`init` line of `file.cio`	No reader. The file is not opened.
`exco_hmet.exc`, `dr_hmet.del`	export coefficient and delivery ratio lines of `file.cio`	Read. Metal loads from routing objects.

Set `metals.mtl`, `hmet_hru.ini` and `hmet_water.ini` to `null` in `file.cio` unless a later revision uses them. The `hmet` columns of `initial.cha` and `initial.aqu` name records of these files and have no effect.

hmet_hru.ini

Initial heavy metals in the soil and on plants of HRUs. The reader expects this layout:

- Line 1: title.
- For each record: a header line, a name line, then for each metal in the order of `constituents.cs` two lines, `label soil` and then `label plant`.

The counting pass expects one line per metal instead of two, so files with more than one record are miscounted. The values (soil and plant concentrations in mg/kg) are stored but no part of the model uses them.

Related files

- `constituents.cs`: the list of metals.
- `file.cio`: the file name slots.
- Pesticides and pathogens: the constituents with full process models.

Sources

- Reader: `hmet_hru_aqu_read.f90`, called from `proc_read.f90`
- Type: `type cs_soil_init_concentrations` (`hmet_soil_ini`) in `constituent_mass_module.f90`
- File name slots: `in_parmdb%hmetcom_db`, `in_init%hmet_soil`, `in_init%hmet_water` in `input_file_module.f90`
- Metal list: `constit_db_read.f90`
- Metal loads: `exco_read_hmet.f90`, `dr_read_hmet.f90`

Item	Fortran variable
hmet_hru.ini record name	`hmet_soil_ini%name`
soil value	`hmet_soil_ini%soil`
plant value	`hmet_soil_ini%plt`
metals.mtl slot	`in_parmdb%hmetcom_db`